

Stratum: illustrated beginner's guide

1. Get to know the workspace

Import a recording, create useful results, and keep every step traceable. Original samples stay unchanged; each operation creates new signals or values.

Start with the example

Launch Stratum and choose **Workspace > Open example workflow**, or choose **Open example workflow** in an empty workspace. The motor-test example loads seven completed steps, from original signals to smoothing, segments and values.

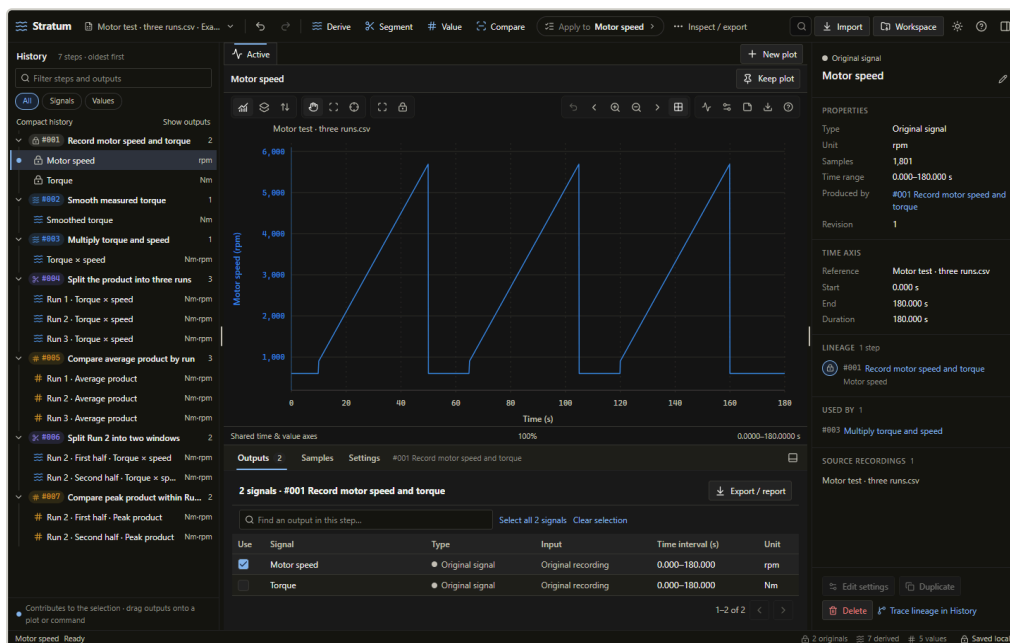


Figure 1. History is on the left and the operations run along the top bar. Active plots the selection above the operation's outputs, and the inspector on the right shows its details and lineage.

Click the arrow beside a step in **History** to expand its outputs, then click a signal name. Type in **Filter steps and outputs**, or choose the **All**, **Signals** or **Values** chip, to find a result; **Ctrl+K** searches every step and output. The recording menu beside **Stratum** switches between recordings.

Import your own data

Choose **Import CSV**. Use a comma-separated UTF-8 file, with time in seconds in the first column and units in square brackets in the header:

```
Time [s],Speed [rpm],Torque [Nm]
0.00,1500,210
0.01,1502,211
0.02,1504,212
```

Times must increase strictly, with no duplicates. Leave missing signal cells empty. Work is saved locally on this device. The screenshots in this guide use the included example, so you can follow along without importing a file.

2. Choose inputs and smooth a signal

The next three pages walk through one small workflow: smooth torque, keep a 10-50 second interval, then calculate its average.

Select the original torque signal

Expand **#001 Record motor speed and torque** and click **Torque**. It appears in **Active**. Check **Apply to** beside the operations in the top bar: it should name **Torque** for this exercise.

Remember: viewing an item and checking processing inputs are separate actions. Checked inputs stay selected while you browse. Choose the **x** on **Apply to** to use the item you are viewing again.

Apply a moving average

Choose **Derive > Filters > Moving average**. Set **Window size** to **5** samples, as shown below. Expand the **review selection** disclosure if you want to check the input name, then choose **Create 1 derived signal**.

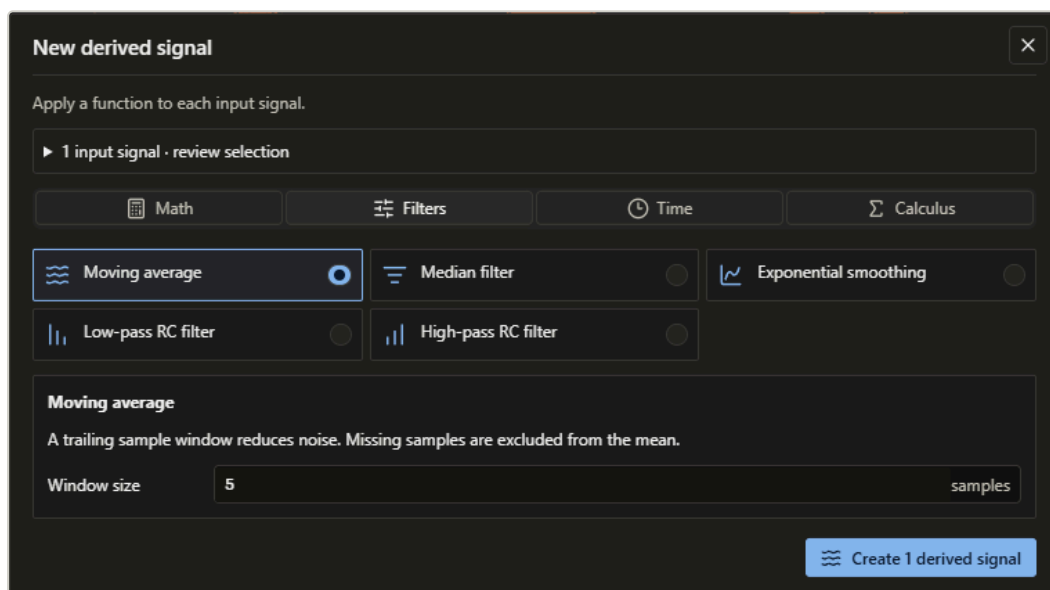


Figure 2. The **Filters** tab contains **Moving average**. **Window size** controls how many samples are used; the button creates a new derived signal.

The new operation appears at the end of history and its output is selected. The original **Torque** signal remains available under step **#001**.

Process several signals together

Select a step: **Active** plots its outputs together, and the **Outputs** tab below the plot lists them. Tick the signals you want to process, then check **Apply to** before opening **Derive**, **Segment** or **Value**. Clear the checked inputs when you want to return to a single selected signal.

3. Keep a useful time interval

A segment is a derived signal covering part of its input. You can process it further or segment it again.

With the new smoothed output selected, choose **Segment > Time ranges**. Click **Add exact range**, set **Start (s)** to **10** and **End (s)** to **50**, then choose **Preview**.

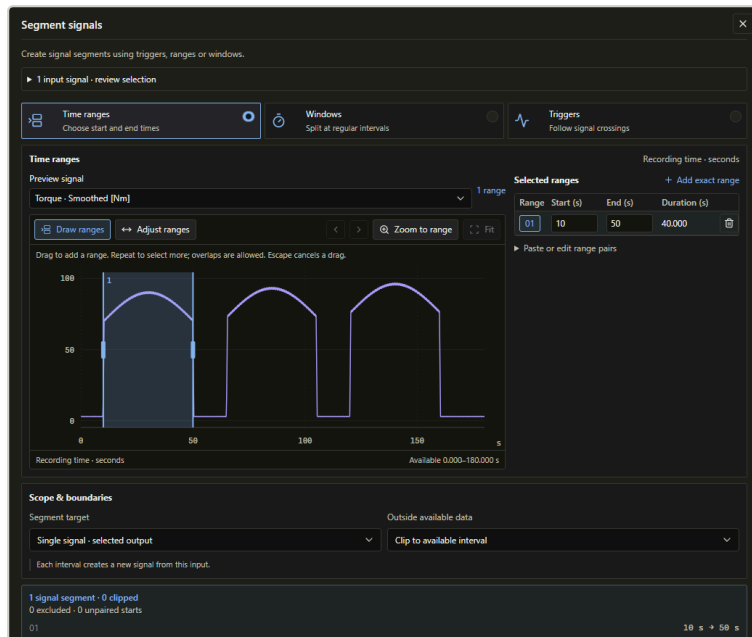


Figure 3. The shaded band and exact fields describe the same interval. The preview confirms one segment from 10 s to 50 s, with no clipped intervals.

Check **Segment target** and the preview boundaries, then choose **Create signal segments**. Scroll down in the dialog if the buttons are below the visible area. Select the new segment in history for the next step.

You can also drag across the plot with **Draw ranges**, or move and resize a range with **Adjust ranges**. **Windows** splits at regular intervals; **Triggers** uses signal threshold crossings. Start with Time ranges while learning the workflow.

4. Turn a signal into a value

Select the segment you just created. Choose **Value > Time average**, review the input, then choose **Create 1 value**.

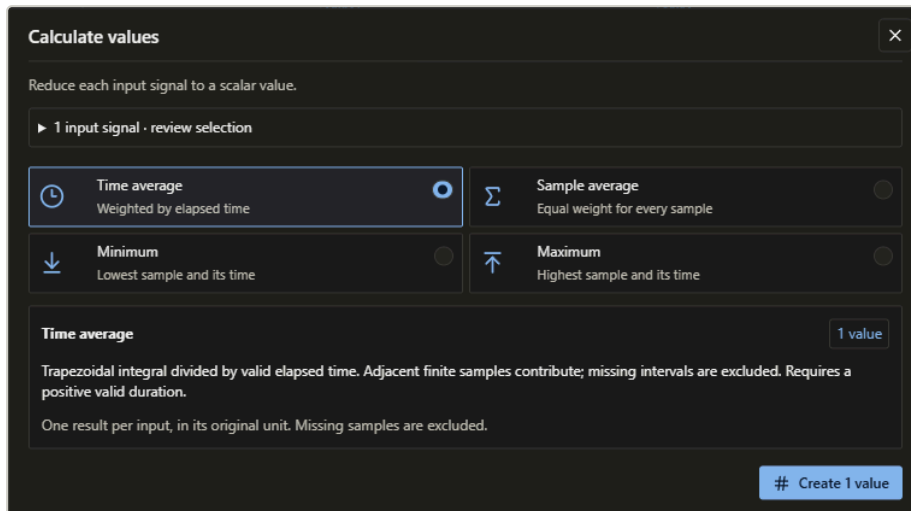


Figure 4. Value creates one number per input signal. The calculation cards explain how each result is calculated.

The new value opens automatically. The result card shows its value, unit, finite sample count and valid time coverage; the plot below draws the value as a labelled dashed line over its input signal.

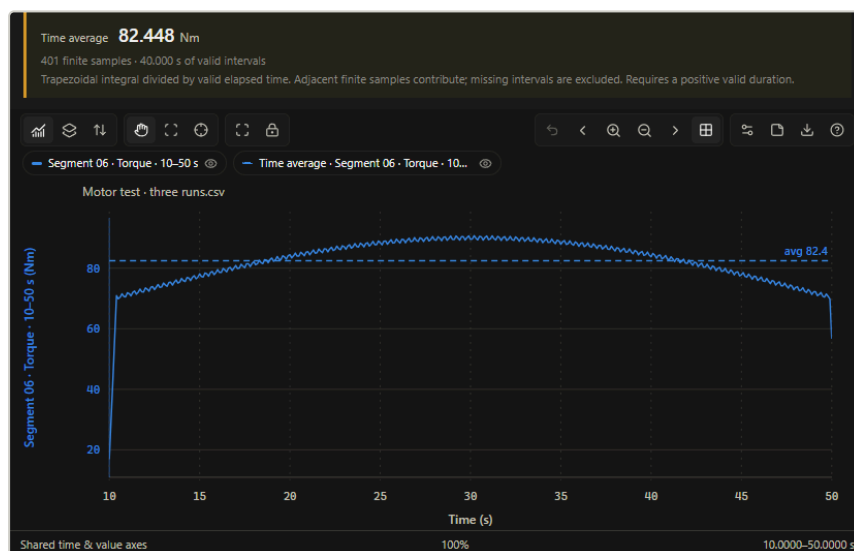


Figure 5. The result card and its reference line from this example exercise. Your own recordings will produce different values and coverage.

Time average weights by elapsed time and excludes missing intervals. **Sample average** gives each finite sample equal weight. **Minimum** and **Maximum** find the lowest and highest finite samples.

Trace and revise your work

The inspector on the right shows the selection's properties, its lineage back to the original recordings and the later operations that use it. **Inspect / export** in the top bar also opens **View samples**, **Inputs and originals**, **Show lineage in tree** and **Used by later operations**. Clear the filter or choose **Show all steps** if History hides something you expect to see.

Use the inspector's buttons, or right-click a History item, to rename it or manage its operation. **Edit settings** updates the operation and recalculates dependent results; **Duplicate** starts a separate branch. **Delete** previews the dependent work it will also remove. **Undo** and **Redo** in the top bar (**Ctrl+Z**, **Ctrl+Y**) keep the last 20 workspace changes, including across restarts.

5. Compare signals on a plot

Active follows the item you inspect. **Keep plot** creates a named plot that stays open while you browse, and **New plot** starts a blank comparison.

Compare the three example runs

Choose **New plot** > **Add signals**. Find the three **Run 1**, **Run 2** and **Run 3** signals named **Torque × speed**, tick them, then choose **Apply signals**. Turn on **Align starts** to compare their elapsed time.

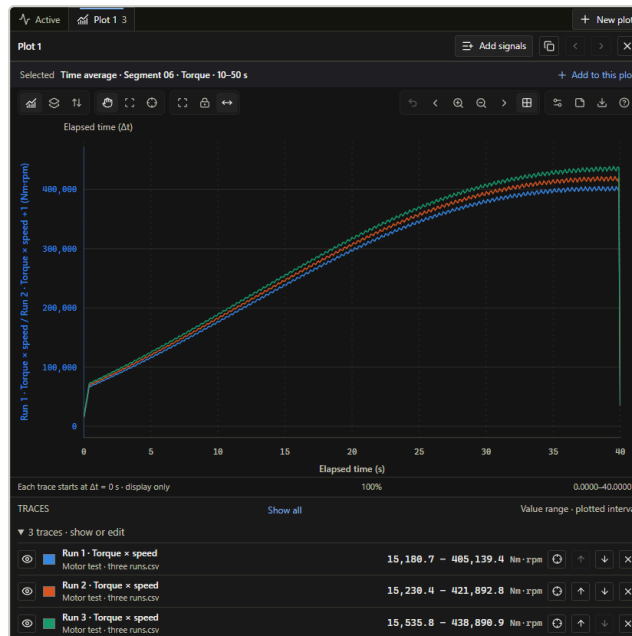


Figure 6. Three runs share an elapsed-time axis. **Align starts** is active; the numbered badge on the plot tab shows that it contains three traces.

Selecting the step that created the runs plots them together on **Active** too. **Align starts** changes the display only. **Compare** opens **Compare & align** when you want to create reusable aligned signals, including signals from different recordings.

Navigate and adjust the comparison

- Click the plot to focus it, then scroll to zoom time. Shift-drag pans in time. Choose **Fit** to see the complete plot again.
- Overlay** draws traces on one time axis, giving different units their own lanes; **Stacked** gives each trace a panel; **Y axes** overlays different units on independent scales. Open the trace list below a plot to show, hide or edit individual traces.
- Drag signals onto a plot to add them. Dragging a segment adds all sibling segments from its operation; **Add signals** lets you choose individual members.
- Use the plot toolbar's **Export** menu to save an SVG or PNG image of the plot.

Named plot layouts are saved on this device, separately from workflow history and workspace backups.

6. Export results and keep a backup

Choose the results to export

Choose **Inspect / export > Export / report**. Set **Include** to the viewed output, checked signals, or all outputs from a step. Choose a **File format**, review the included outputs, then choose **Download file**.

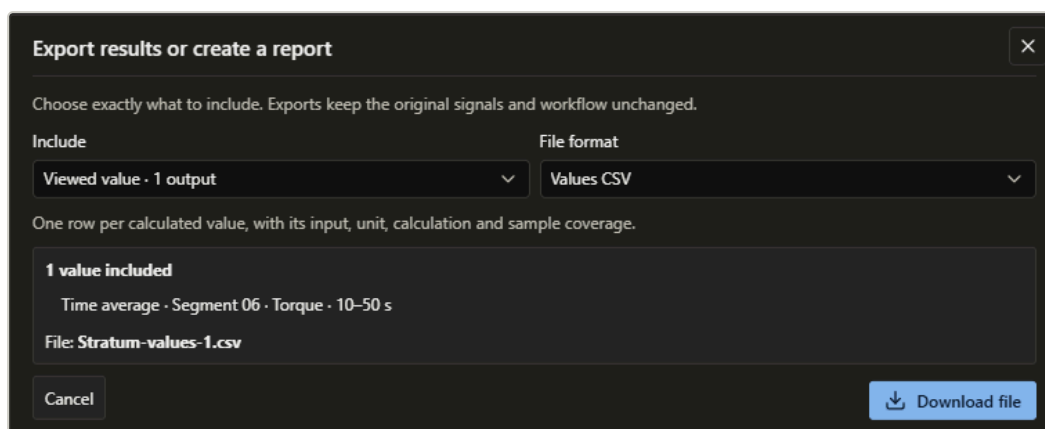


Figure 7. This export contains just the viewed value. Check Include before downloading so you get the intended output or batch.

Format	Use it for
Samples CSV	Every evaluated sample, with its time axis; missing values stay blank.
Values CSV	One row per calculated value, with units and its input.
Signal summary CSV	A compact summary of each signal, without individual samples.
Printable report (HTML)	A snapshot of results, plots and contributing history. Open in a browser to print or save as PDF.

Save the whole workspace

Choose **Workspace > Download workspace backup** and keep the **.stratum** file somewhere safe. Older **.stratus** backups can still be restored.

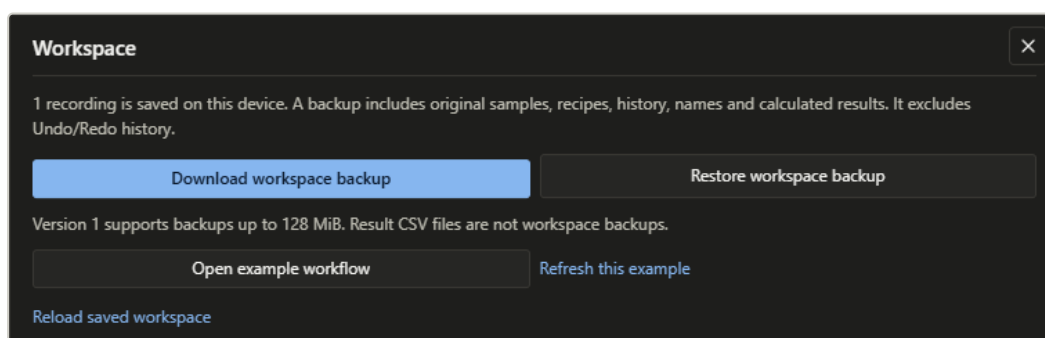


Figure 8. Workspace backups preserve the analysis for later restoration. Result CSV files are not workspace backups.

A backup includes original samples, recipes, results, names and history; saved plot layouts and Undo/Redo history are excluded. **Restore workspace backup** replaces the current workspace after confirmation; Undo can recover the prior one.

Desktop and browser workspaces have separate local storage. Use a backup to move your analysis between them or to another computer. Backups support up to **128 MiB**; Samples CSV supports **64 MiB** per file. For a larger sample export, choose fewer signals or shorter segments. The **?** button in the top bar opens the built-in workflow help, and the sun or moon button switches between light and dark themes.